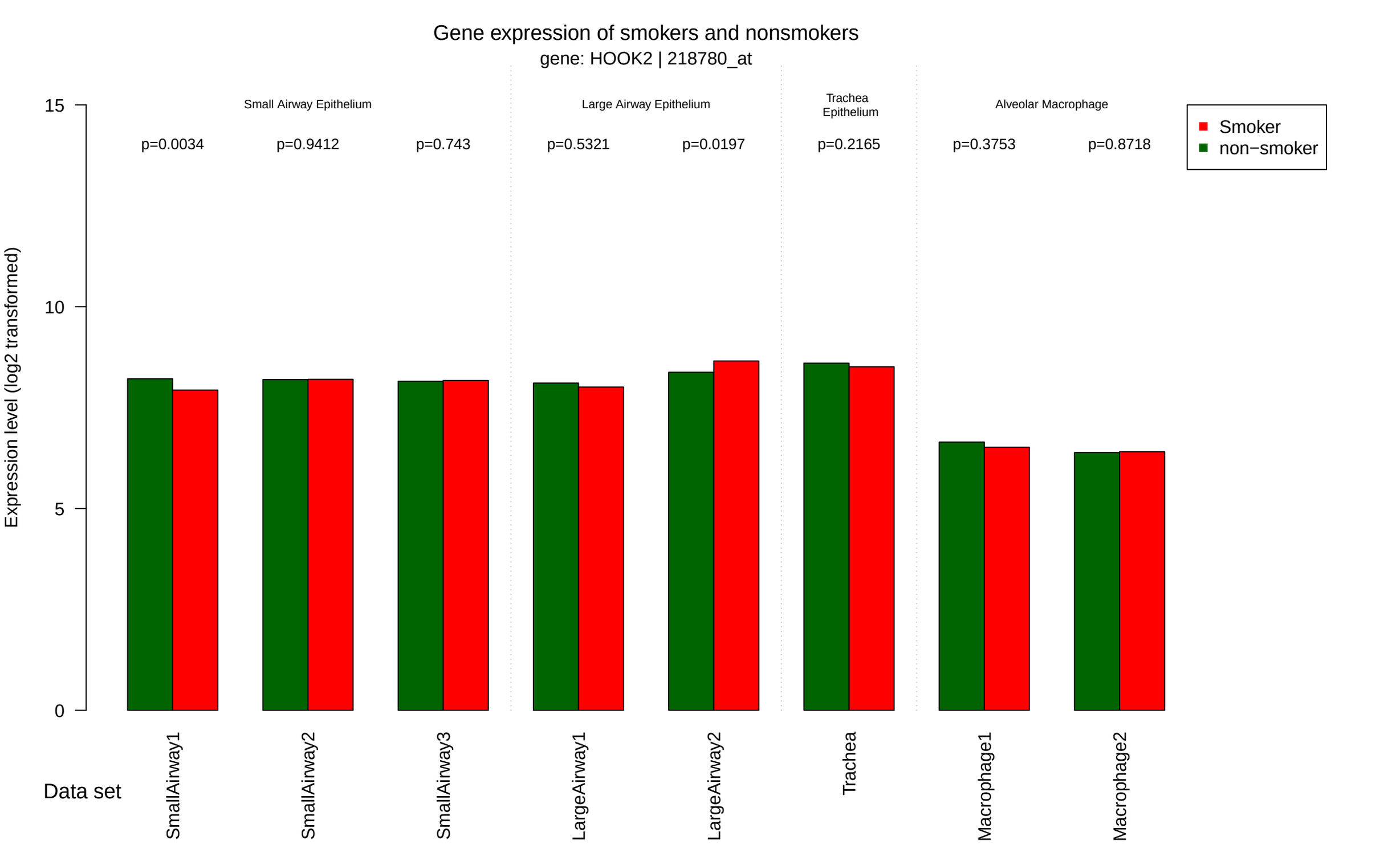
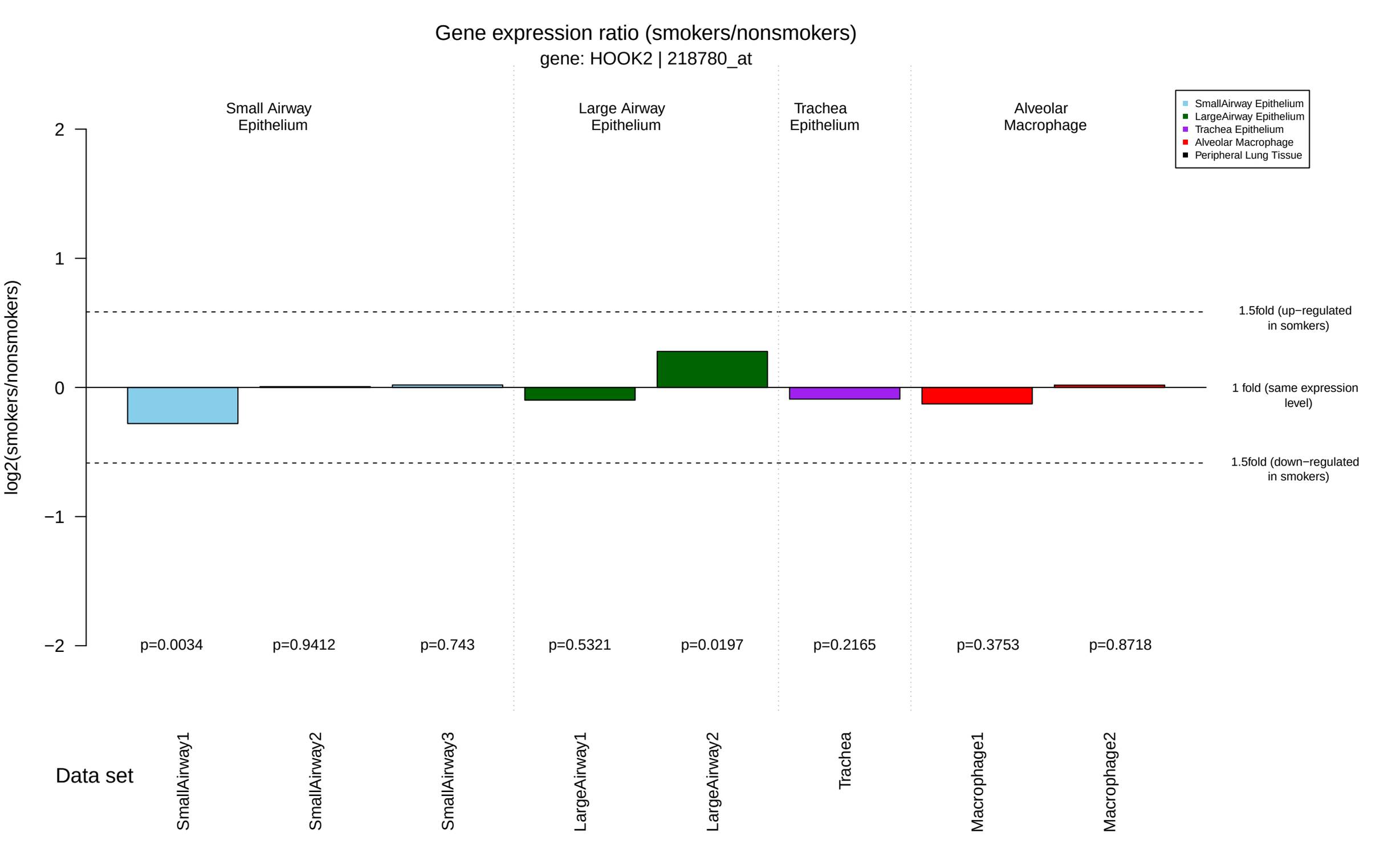
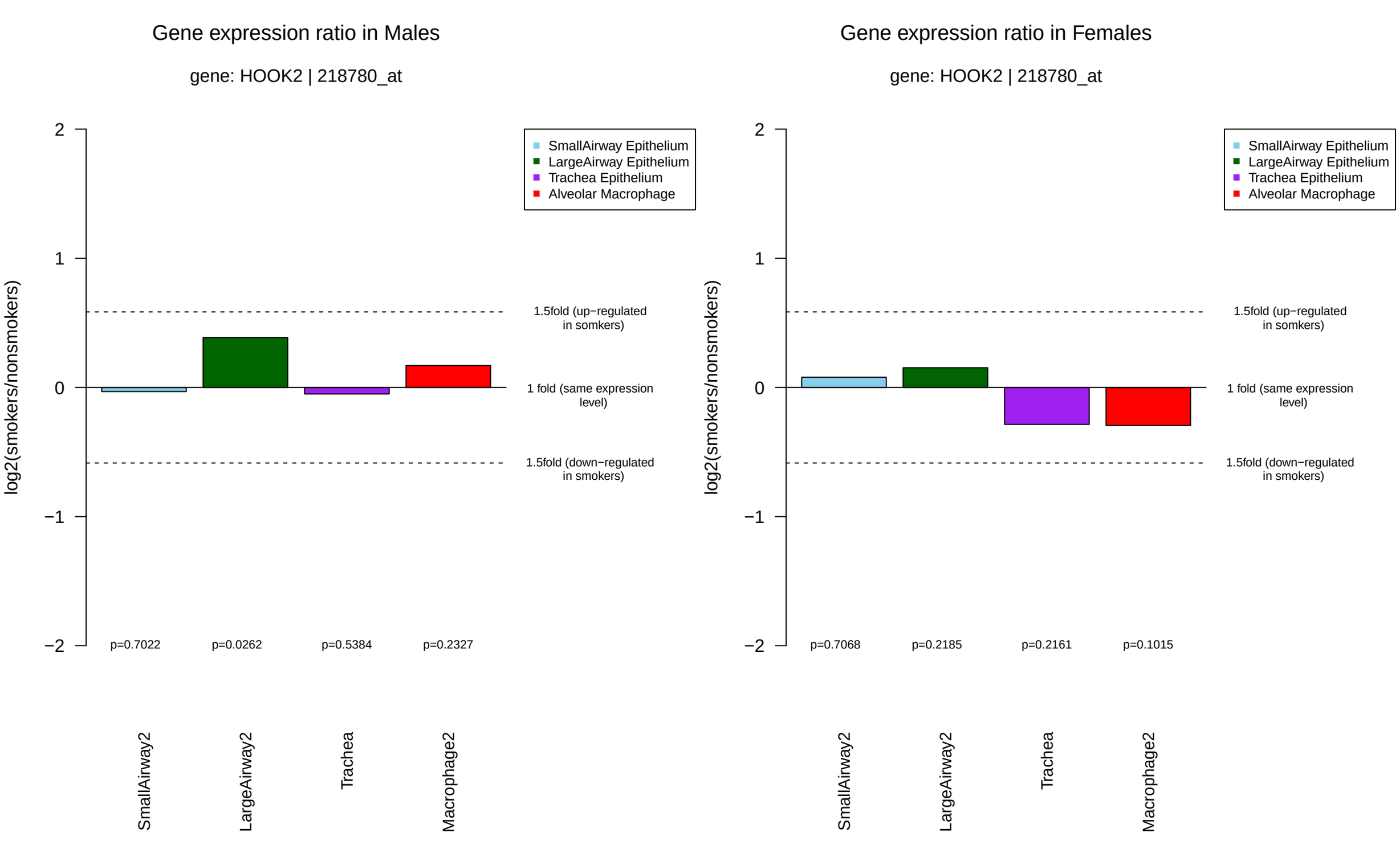
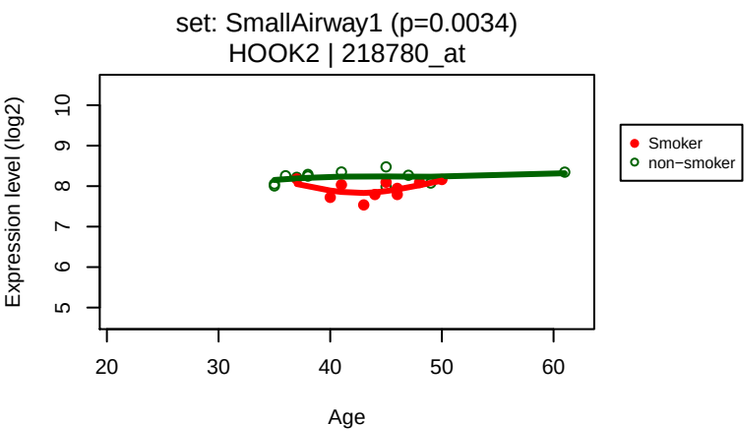
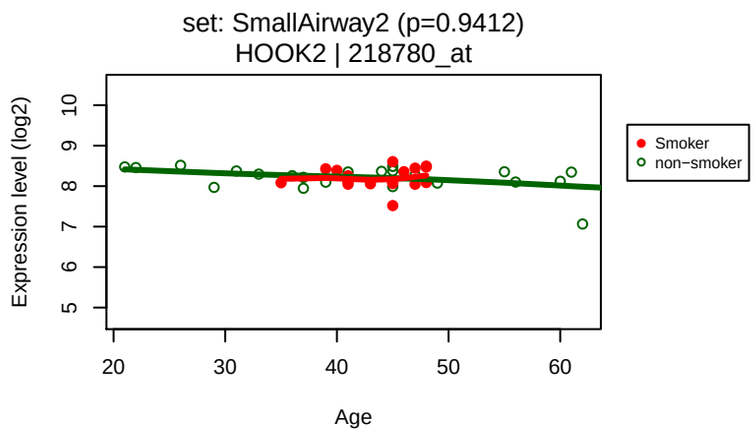
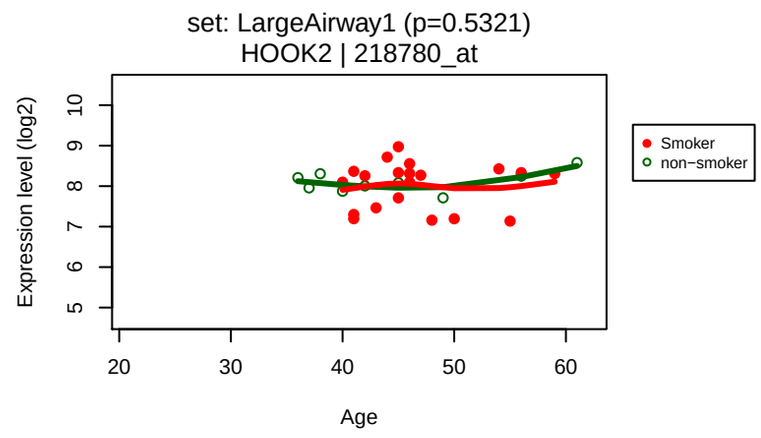

Gene expression of smokers and nonsmokers
gene: HOOK2 | 218780_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0034
p=0.9412
p=0.743
p=0.5321
p=0.0197
p=0.2165
p=0.3753
p=0.8718
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

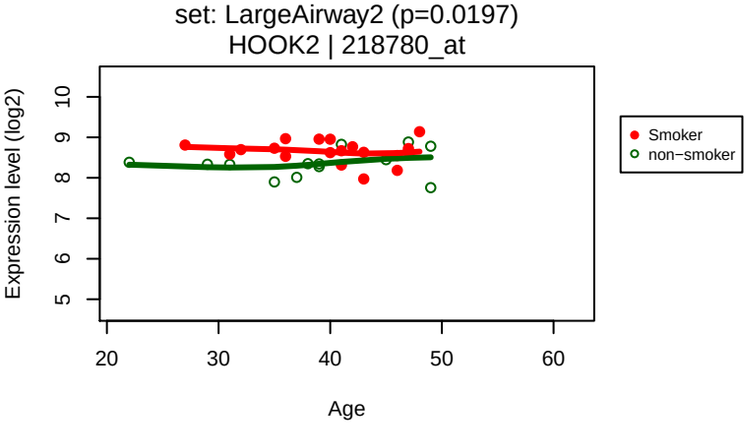

Gene expression ratio (smokers/nonsmokers)
gene: HOOK2 | 218780_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0034
p=0.9412
p=0.743
p=0.5321
p=0.0197
p=0.2165
p=0.3753
p=0.8718
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

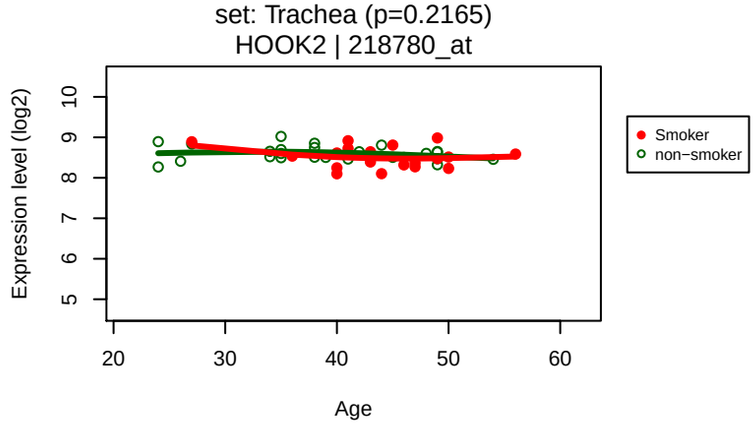

Gene expression ratio in Males
gene: HOOK2 | 218780_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7022
p=0.0262
p=0.5384
p=0.2327
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOOK2 | 218780_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7068
p=0.2185
p=0.2161
p=0.1015
SmallAirway2
LargeAirway2
Trachea
Macrophage2

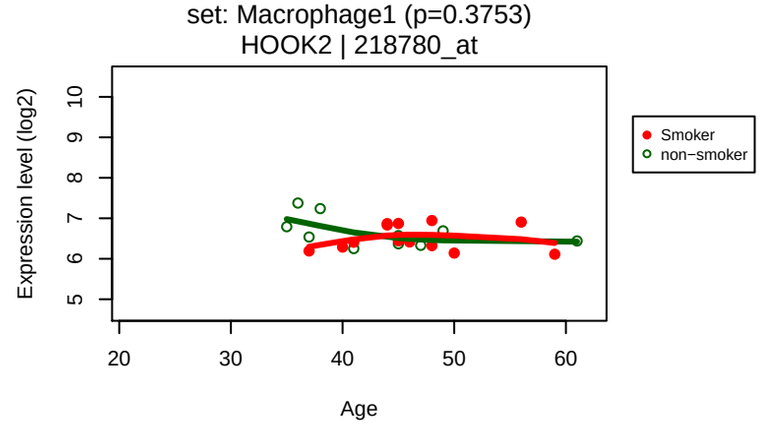

set: SmallAirway1 (p=0.0034)
HOOK2 | 218780_at
Expression level (log2)
Age
Smoker
non−smoker

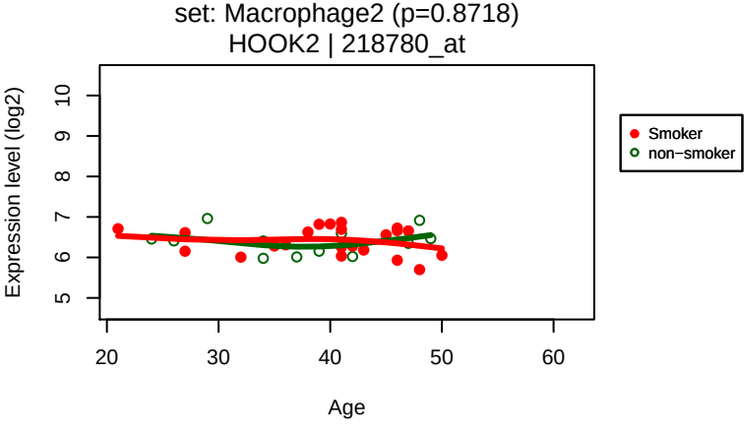

set: SmallAirway2 (p=0.9412)
HOOK2 | 218780_at
Expression level (log2)
Age
Smoker
non−smoker

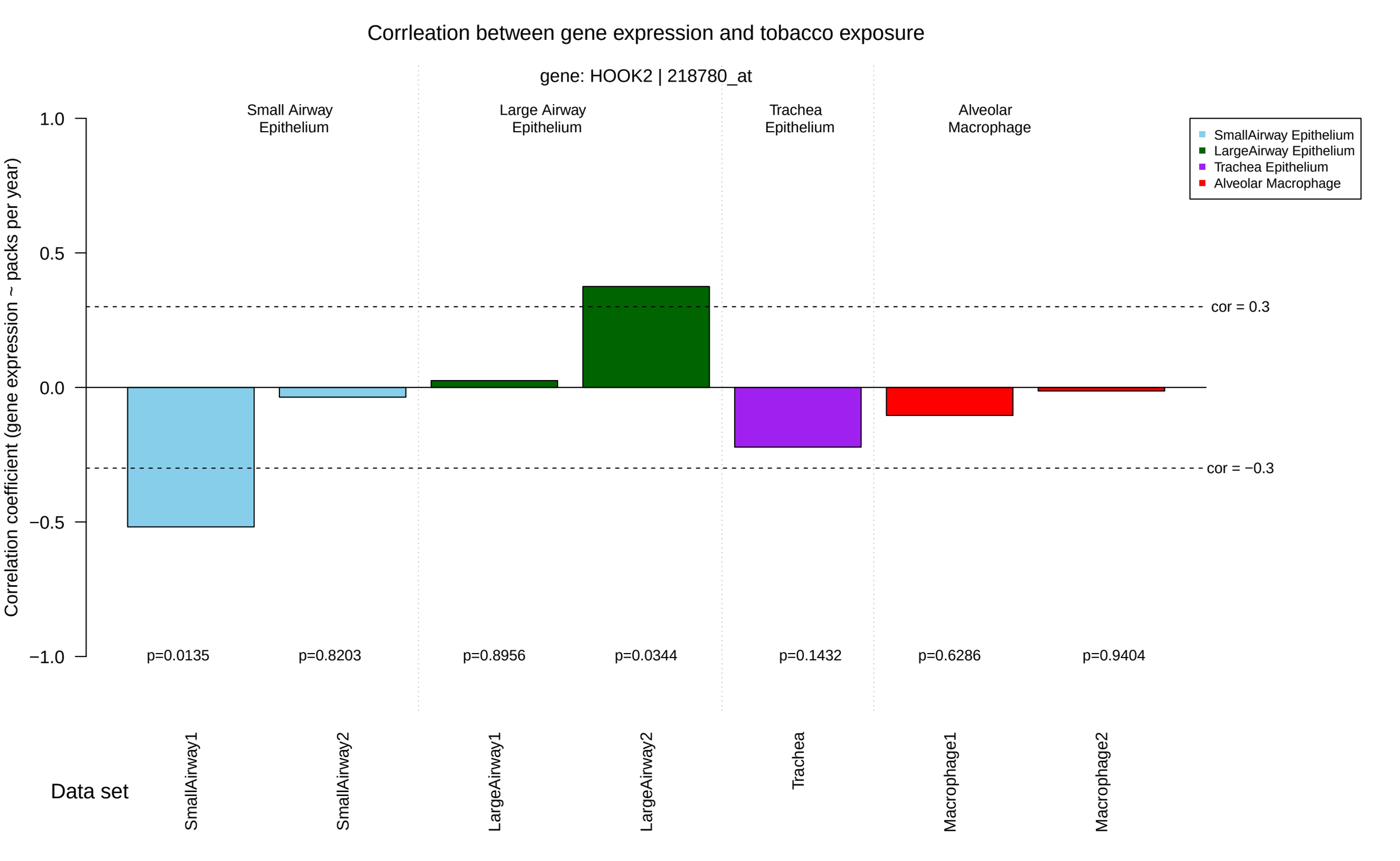

set: LargeAirway1 (p=0.5321)
HOOK2 | 218780_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0197)
HOOK2 | 218780_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2165)
HOOK2 | 218780_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3753)
HOOK2 | 218780_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8718)
HOOK2 | 218780_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOOK2 | 218780_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0135
p=0.8203
p=0.8956
p=0.0344
p=0.1432
p=0.6286
p=0.9404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2